Mylan		
	Document Name	Storage Migration using NPIV
	Document Number	MYL-UNX-SOP-SMNPIV- 001
	Type of document	Standard Operation Procedure
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1. PURPOSE

1.1 This document defines the NIPV configuration on individual LPARs and Storage Migration procedures.

2. SCOPE

2.1 As part of Storage Migration, we are proposing NPIV configuration to reduce the VIO IO handle Issues.

2.2 We need to create NPIV adapter in HMC on LPAR profile.

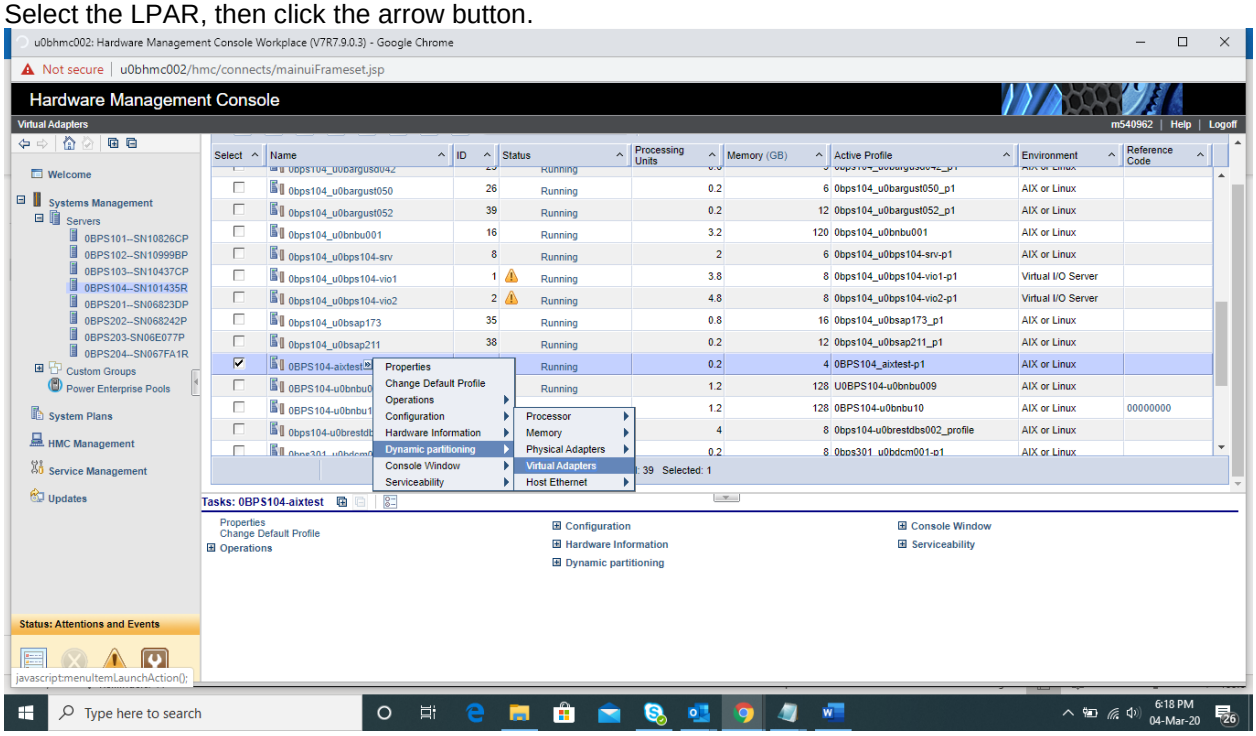
2.3 We should map those adapters to VIO Adapter, then configure it on LPAR.

2.4 Storage team will map the PowerMax LUNs to NPIV adapter, UNIX Team will use this LUNs for storage migration

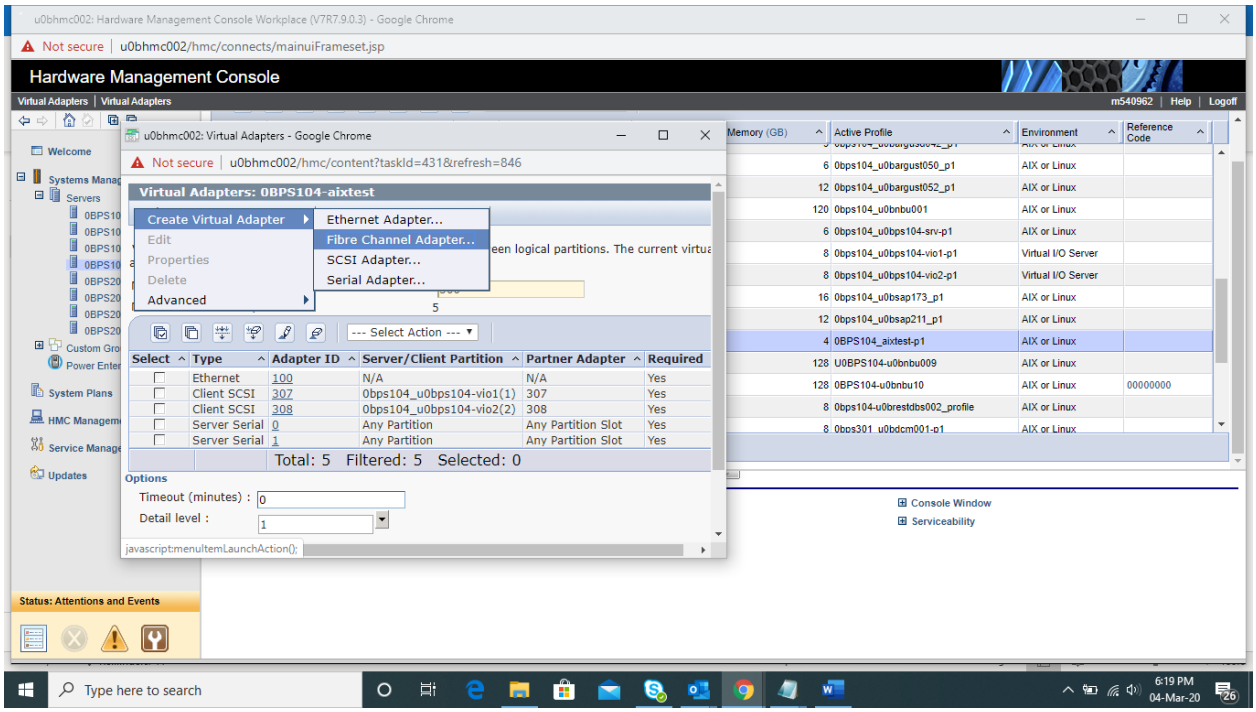
3. PROCEDURE:

3.1. NPIV Configuration

1. We should login into the HMC using our MID and password
2. Expand the Server Management, the Navigate to Servers.
3. Expand the Servers, and click on the appropriate Physical Frames
4. Select the LPAR, then click the arrow button.



5. Navigate to Dynamic partitioning, then click on virtual adapter
6. Now virtual adapter Popup Screen will open.
7. Click Action Button, Navigate to Create virtual adapter, then click the fiber channel adapter

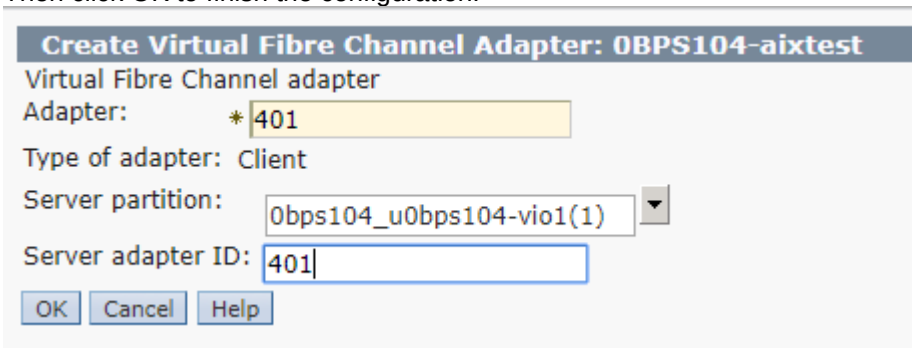


8. Create Virtual Adapter Popup will open, there we should define the Adapter ID
9. Server Partition filed, we should select the VIO, the we need to define the VIO adapter ID

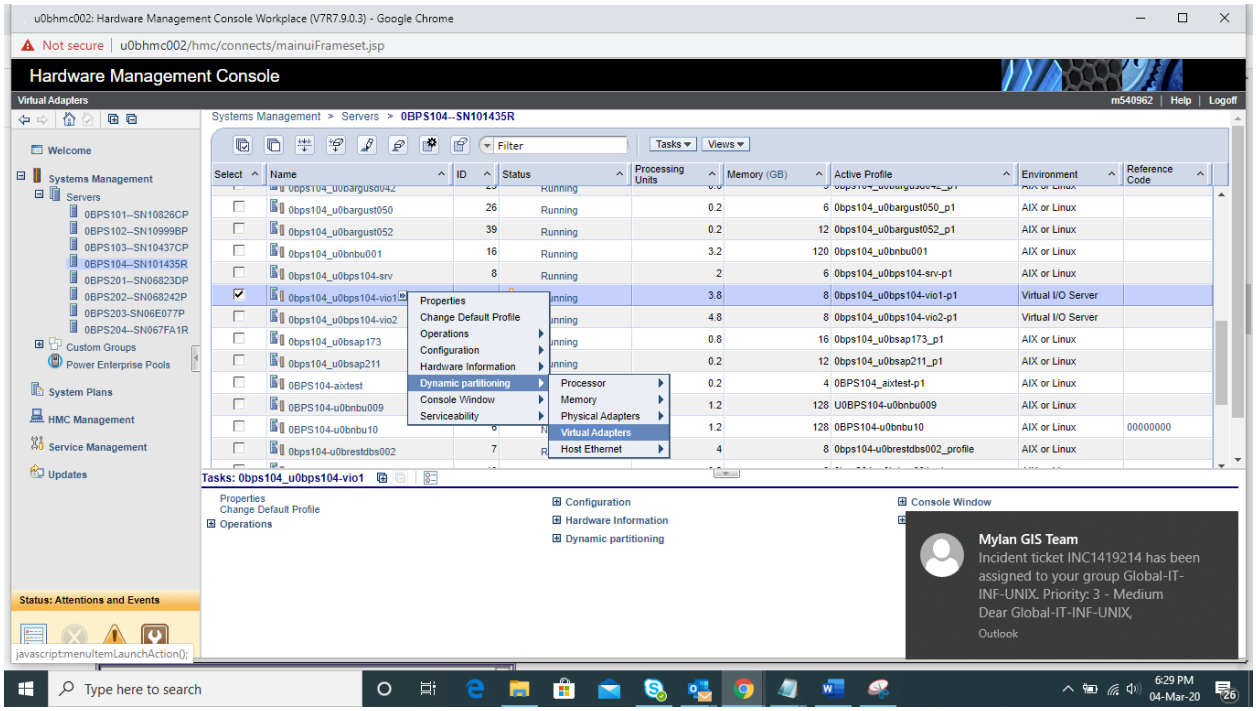
   We should use the below numeric values for virtual adapters. if lpar ID 10 then adaptor id will be 106 at client virtual adapter, And same ID should be create in VIO as well.

Standards for Adapters		
Ethernet Adapter	**vscsi Adapter**	**FC Adapter**
0,1	2,3,4,5	6,7

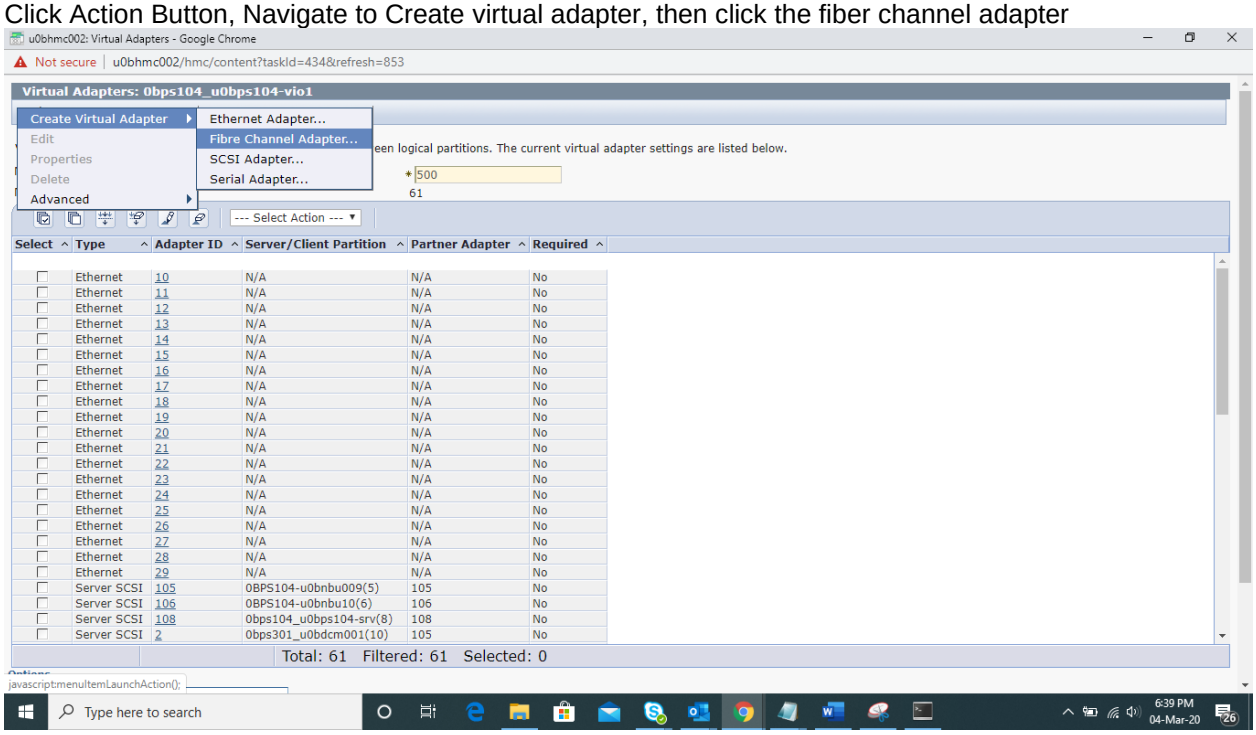
10. Then click OK to finish the configuration.



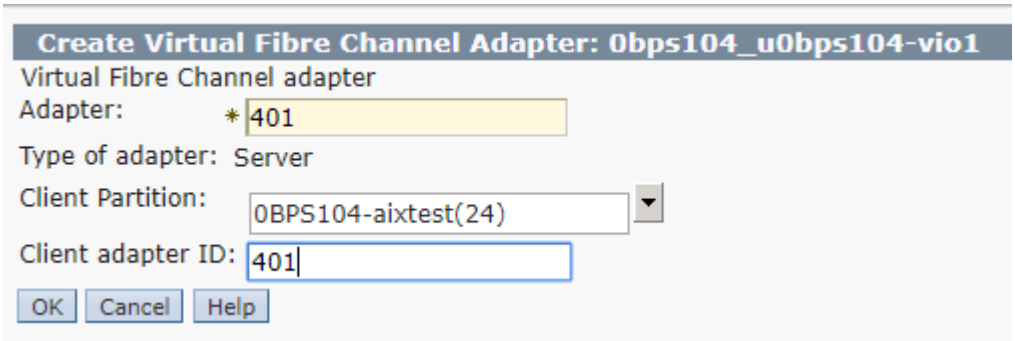
11. Now select the VIO, click on Arrow button
12. Navigate to Dynamic Adapter, then click on virtual adapter



13. Now virtual adapter Popup Screen will open.
14. Click Action Button, Navigate to Create virtual adapter, then click the fiber channel adapter

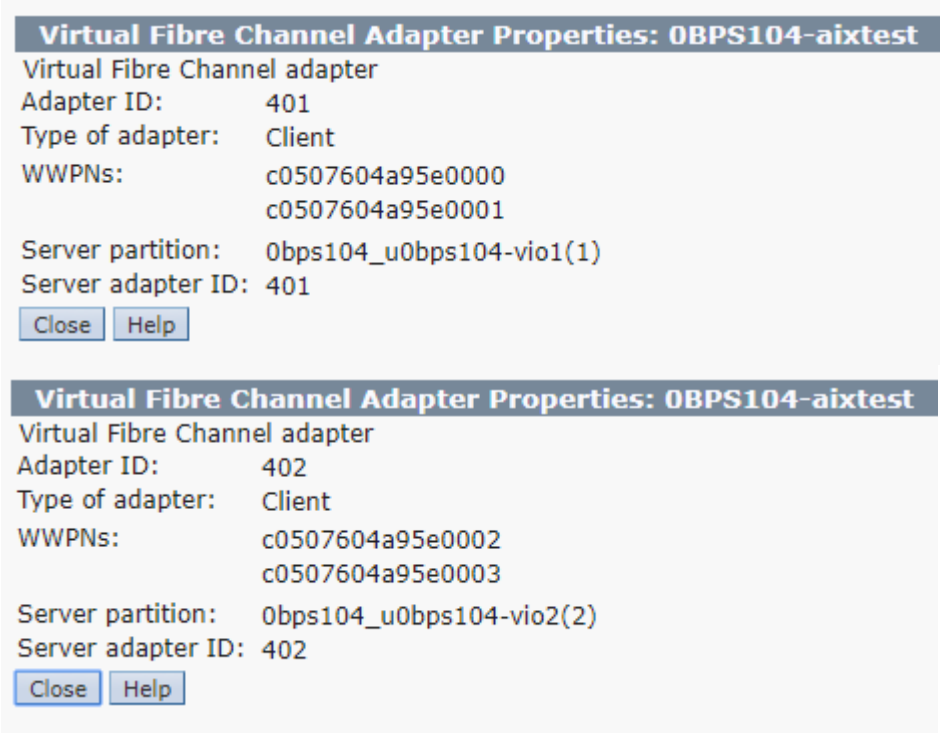


15. Create Virtual Adapter Popup will open, there we should define the Adapter ID
16. Client Partition filed, we should select the Appropriate LPAR, the we need to give LPAR adapter ID, which we gave in the step – 8.
17. Then click OK to finish the configuration.



Note: we have completed the Adapter creation on VIO1 and LPAR, for reduntancy purpose, we need to create it on VIO2 so repeat the step 4 to step14

18. Validate the WWPN details LPAR virtual adapter properties



Note: If any error happened on rmc connection, use below mentioned command for start the service.

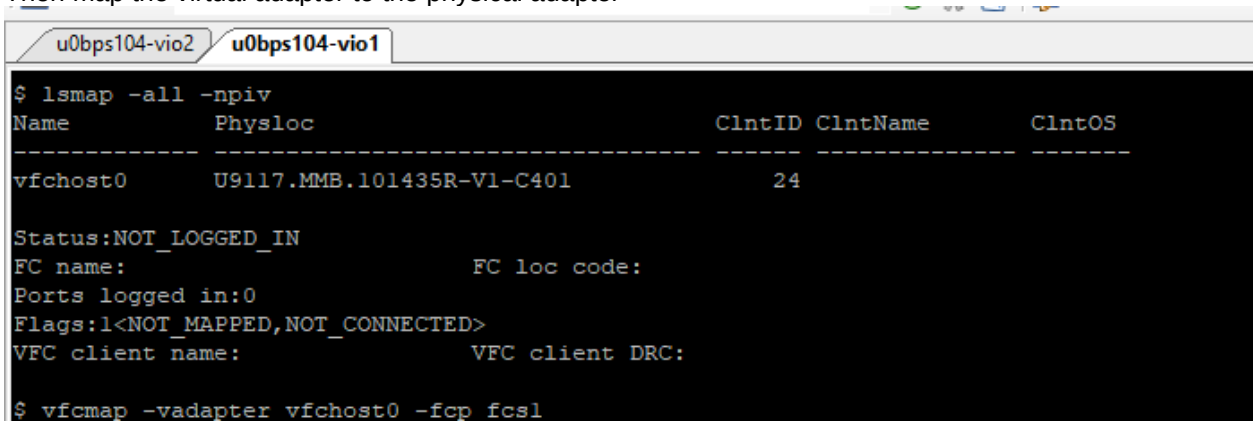
```
# /usr/sbin/rsct/bin/rmcctrl -s*C
# /usr/sbin/rsct/bin/rmcdomainstatus -s ctrmc

Management Domain Status: Management Control Points
  I A  0xc538213dc2ea42fc  0001  172.20.26.75
# /usr/sbin/rsct/bin/rmcctrl -z
# /usr/sbin/rsct/bin/rmcctrl -A
0513-059 The ctrmc Subsystem has been started. Subsystem PID is 8061074.
# /usr/sbin/rsct/bin/rmcctrl -p
```

19. We should map the adapters in VIO level, hence we need to login into the VIO's
20. To discover the device on VIO, run "cfgdev" command.



21. Then check the NPIV Adapter details in VIOs
22. Using lsnports command check the active physical adapter.
23. Then map the virtual adapter to the physical adapter



u0bps104-vio2 u0bps104-vio1

```
$ lsmap -all -npiv
Name          Physloc                            ClntID ClntName       ClntOS
------------- ---------------------------------- ------ -------------- -------
vfchost0      U9117.MMB.101435R-V2-C402               24

Status:NOT_LOGGED_IN
FC name:                      FC loc code:
Ports logged in:0
Flags:1<NOT_MAPPED,NOT_CONNECTED>
VFC client name:              VFC client DRC:

$ vfcmap -vadapter vfchost0 -fcp fcs1
$ lsmap -all -npiv
Name          Physloc                            ClntID ClntName       ClntOS
------------- ---------------------------------- ------ -------------- -------
vfchost0      U9117.MMB.101435R-V2-C402               24

Status:NOT_LOGGED_IN
FC name:fcs1                  FC loc code:U78C0.001.DBJO974-P2-C1-T2
Ports logged in:0
Flags:4<NOT_LOGGED>
VFC client name:              VFC client DRC:

$
```

u0bps104-vio2 u0bps104-vio1

```
$ lsmap -all -npiv
Name          Physloc                            ClntID ClntName       ClntOS
------------- ---------------------------------- ------ -------------- -------
vfchost0      U9117.MMB.101435R-V1-C401               24

Status:NOT_LOGGED_IN
FC name:fcs1                  FC loc code:U78C0.001.DBJM826-P2-C1-T2
Ports logged in:0
Flags:4<NOT_LOGGED>
VFC client name:              VFC client DRC:
```

24. Now login to the LPAR, then switch to root “sudo su –“
25. Discover the devices, run the cfgmgr command

uname -n

u0bps201-srv

lsdev -Cc adapter

ent0 Available Virtual I/O Ethernet Adapter (l-lan)

vsa0 Available LPAR Virtual Serial Adapter

vscsi0 Available Virtual SCSI Client Adapter

vscsi1 Available Virtual SCSI Client Adapter

cfgmgr

lsdev -Cc adapter

ent0 Available Virtual I/O Ethernet Adapter (l-lan)

fcs0 Available 22-T1 Virtual Fibre Channel Client Adapter

vsa0 Available LPAR Virtual Serial Adapter

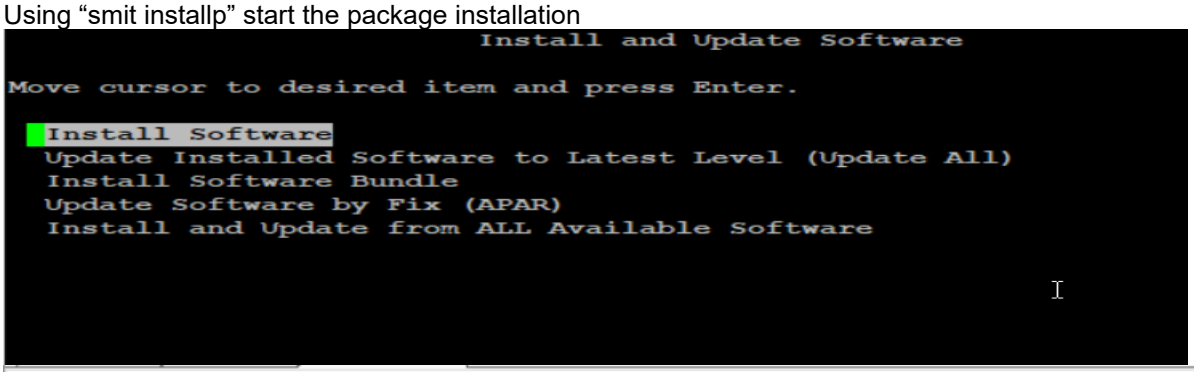
vscsi0 Available Virtual SCSI Client Adapter

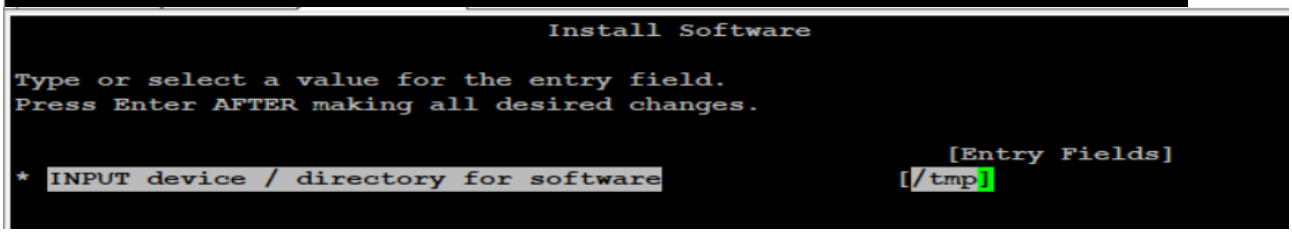
vscsi1 Available Virtual SCSI Client Adapter

#

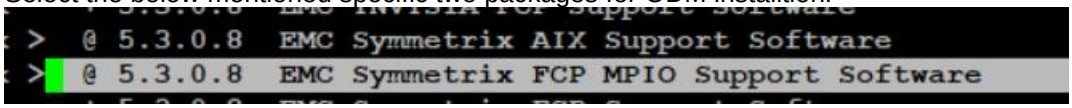
3.2 Storage Migration

26. Provide the WWPN to storage team for allocate the LUN, they will give you the LUNs from PowerMax Storage.
27. Before scan the LUN from LPAR client, we should install the EMC ODM package.
28. For AIX 5.3 and 6.1, we need to use “EMC.AIX.5.3.0.8.tar” package and Aix 7.1 we nned to use “EMC.AIX.6.0.0.0.tar”
29. Extract the file in /tmp directory.
30. Using “smit installp” start the package installation





31. Select the below mentioned specific two packages for ODM installtion.



32. Once Installtion completed, Storage team will provide the LUN, scan it from LPAR level. Using cfgmgr command.
33. Check the newly added disk in “lspv | grep -I none” and check the created time on “ls -lrt /dev/hdisk*”
34. Use the attr_disk.ksh script for change the disk attributes. “/usr/admin/aix/scripts/pmax_mig/attr_disk.ksh” It will modify the below mentioned attributes **“algorithm round_robin, queue_depth 32, reserve_policy no_reserve, max_transfer 0x100000”**
35. Add the PVID on newly added disks.

root@u0bps201-srv /> chdev -l hdisk1 -a pv=yes

hdisk1 changed

root@u0bps201-srv /> lspv

hdisk0 00f6823d11048fa5 rootvg active

hdisk1 00f6823ded175094 None

36. We should add the PV in to Existing VG via extendvg, then check the PP size in pv and vg level. Then we have to migrate the data from old disk to new disk by start the mirroring. “command is mirrorvg, Once Mirroring is completed. Kindly validate the lv state and pv occupation size of new disk. If everything migrated, then we need to unmirror the volume from the VG, the remove the disk from the vg using reduce vg.

root@u0bps201-srv /> extendvg rootvg hdisk1
root@u0bps201-srv /> mirrorvg -S rootvg hdisk1
0516-1259 mirrorvg: None of the available disks support booting on this
specific system or the boot logical volume already exists on
the specified disk.
If these are new bootable disks and you
have not rebooted since configuring them. You may try
exporting LVM_HOTSWAP_BOOTDISK=1 and run this command again
to override this condition.
0516-1200 mirrorvg: Failed to mirror the volume group.
root@u0bps201-srv /> mirrorvg -S rootvg hdisk1
0516-1259 mirrorvg: None of the available disks support booting on this
specific system or the boot logical volume already exists on
the specified disk.
If these are new bootable disks and you
have not rebooted since configuring them. You may try
exporting LVM_HOTSWAP_BOOTDISK=1 and run this command again
to override this condition.
0516-1200 mirrorvg: Failed to mirror the volume group.
root@u0bps201-srv /> export LVM_HOTSWAP_BOOTDISK=1
root@u0bps201-srv /> mirrorvg -S rootvg hdisk1
0516-1804 chvg: The quorum change takes effect immediately.
0516-1126 mirrorvg: rootvg successfully mirrored, user should perform
bosboot of system to initialize boot records. Then, user must modify
bootlist to include: hdisk0 hdisk1.
root@u0bps201-srv /> lspv hdisk1
PHYSICAL VOLUME: hdisk1 VOLUME GROUP: rootvg
PV IDENTIFIER: 00f6823ded175094 VG IDENTIFIER 00f6823d00004c000000015f15c94bdb
PV STATE: active
STALE PARTITIONS: 190 ALLOCATABLE: yes
PP SIZE: 128 megabyte(s) LOGICAL VOLUMES: 11
TOTAL PPs: 479 (61312 megabytes) VG DESCRIPTORS: 1
FREE PPs: 283 (36224 megabytes) HOT SPARE: no
USED PPs: 196 (25088 megabytes) MAX REQUEST: 512 kilobytes
FREE DISTRIBUTION: 95..03..00..89..96
USED DISTRIBUTION: 01..93..95..07..00
MIRROR POOL: None
root@u0bps201-srv /> lsvg rootvg
VOLUME GROUP: rootvg VG IDENTIFIER: 00f6823d00004c000000015f15c94bdb
VG STATE: active PP SIZE: 128 megabyte(s)

VG PERMISSION: read/write TOTAL PPs: 863 (110464 megabytes)
MAX LVs: 256 FREE PPs: 471 (60288 megabytes)
LVs: 11 USED PPs: 392 (50176 megabytes)
OPEN LVs: 10 QUORUM: 1 (Disabled)
TOTAL PVs: 2 VG DESCRIPTORS: 3
STALE PVs: 1 STALE PPs: 86
ACTIVE PVs: 2 AUTO ON: no
MAX PPs per VG: 32512
MAX PPs per PV: 1016 MAX PVs: 32
LTG size (Dynamic): 512 kilobyte(s) AUTO SYNC: no
HOT SPARE: no BB POLICY: relocatable
PV RESTRICTION: none INFINITE RETRY: no
DISK BLOCK SIZE: 512 CRITICAL VG: no
FS SYNC OPTION: no

37. Install the boot Image on the new disk. Then remove the old disk from LPAR and VIO level.

root@u0bps201-srv /> bosboot -ad /dev/hdisk1
trustchk: /etc/vfs: Verification of attributes failed: mode

bosboot: Boot image is 51228 512 byte blocks.
root@u0bps201-srv /> bootlist -m normal -o hdisk1 hdisk0
hdisk1 blv=hd5 pathid=0
hdisk1 blv=hd5 pathid=1
hdisk0 blv=hd5 pathid=0
hdisk0 blv=hd5 pathid=1
root@u0bps201-srv /> bootlist -m normal -o
hdisk1 blv=hd5 pathid=0
hdisk1 blv=hd5 pathid=1
hdisk0 blv=hd5 pathid=0
hdisk0 blv=hd5 pathid=1
root@u0bps201-srv /> unmirrorvg rootvg hdisk0
0516-1246 rmlvcopy: If hd5 is the boot logical volume, please run 'chpv -c <diskname>'
as root user to clear the boot record and avoid a potential boot
off an old boot image that may reside on the disk from which this
logical volume is moved/removed.

0516-1804 chvg: The quorum change takes effect immediately.
0516-1144 unmirrorvg: rootvg successfully unmirrored, user should perform
bosboot of system to reinitialize boot records. Then, user must modify
bootlist to just include: hdisk1.
root@u0bps201-srv />
root@u0bps201-srv /> chpv -c hdisk0
root@u0bps201-srv /> bootlist -m normal -o
hdisk1 blv=hd5 pathid=0
hdisk1 blv=hd5 pathid=1
hdisk0 pathid=0
hdisk0 pathid=1

```
root@u0bps201-srv /> ipl_var
root@u0bps201-srv /> ipl_varon -i
ksh: ipl_varon:  not found
root@u0bps201-srv /> ipl_varyon -i
[S 14417964 16515210 03/18/20-06:28:21:789 ipl_varyon.c 1313] ipl_varyon -i
PVNAME          BOOT DEVICE     PVID                    VOLUME GROUP ID
hdisk0          NO              00f6823d11048fa50000000000000000        00f6823d00004c00
hdisk1          YES             00f6823ded1750940000000000000000        00f6823d00004c00
[E 14417964 0:593 ipl_varyon.c 1454] ipl_varyon: exited with rc=0
root@u0bps201-srv /> reducevg rootvg hdisk0
root@u0bps201-srv /> rmdev -dl /dev/hdisk0
rmdev: 0514-519 The following device was not found in the customized
device configuration database:
name = '/dev/hdisk0'
root@u0bps201-srv /> rmdev -dl hdisk0
hdisk0 deleted
root@u0bps201-srv /> lspv
hdisk1          00f6823ded175094                    rootvg          active
root@u0bps201-srv /> oslevel -s
7100-04-03-1642
root@u0bps201-srv />
root@u0bps201-srv />
root@u0bps201-srv />
root@u0bps201-srv />
root@u0bps201-srv /> shutdown -Fr now

SHUTDOWN PROGRAM
Wed Mar 18 06:38:01 EDT 2020
Running /etc/rc.d/rc2.d/K45vxpbx_exchanged stop
Stopped Symantec Private Branch Exchange
Running /etc/rc.d/rc2.d/KBESClientd stop
Stopping BESClient
Running /etc/rc.d/rc2.d/KrsctPHA stop
Running /etc/rc.d/rc2.d/Ksshd stop
0513-044 The sshd Subsystem was requested to stop.
Running /etc/rc.d/rc2.d/Kwpars stop
Wait for 'Rebooting...' before stopping.
Error logging stopped...
Advanced Accounting has stopped...
Process accounting stopped...
Stopping NFS/NIS Daemons
0513-004 The Subsystem or Group, nfsd, is currently inoperative.
0513-044 The biod Subsystem was requested to stop.
0513-044 The rpc.lockd Subsystem was requested to stop.
0513-044 The rpc.statd Subsystem was requested to stop.
0513-004 The Subsystem or Group, gssd, is currently inoperative.
```

0513-004 The Subsystem or Group, nfsrgyd, is currently inoperative.

0513-004 The Subsystem or Group, rpc.mountd, is currently inoperative.

0513-004 The Subsystem or Group, ypbind, is currently inoperative.

Stopping TCP/IP daemons: ndpd-host lpd routed gated sendmail inetd named timed rwhod iptrace dpid2 snmpd rshd rlogind telnetd syslogd

Removing TCP/IP lock files

VIO:

lspv | grep 00f6823d11048fa5

hdisk8 00f6823d11048fa5 None

exit

$ lsmap -all|grep -ip hdisk8

VTD 0b201srvX272FT

Status Available

LUN 0x8200000000000000

Backing device hdisk8

Physloc U78AA.001.WZSGB3E-P1-C4-T1-W50017380EAAB0142-L0

Mirrored false

$ rmvdev -vtd 0b201srvX272FT

0b201srvX272FT deleted

$

As per the PowerMax Storage vendor, reboot is not required post migration. But they are asking us to reboot the server post ODM package installation.

4. DEFINITIONS:

LPAR & VIO	AIX logical partition & IBM Virtual IO Server
NPIV	N_Port ID Virtualization used in the SAN virtualization
HMC	Hardware management Console
MYL-UNX-SOP-SMNPIV- 001	Storage Migration with NPIV configuration

5. VERSION HISTORY

	Prepared By	Reviewed by	Approved By
Name	Vinoth Sankar	Franklin K Dhas	Arunachalam G
Role	SME	SME	Senior Manager
Date	**07-APR-2020**	**23-Oct-2020**	**27-Oct-2020**
Version	**1.0**	**1.0**	**1.0**